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## LIQUID STORAGE BOTTLE AND RECORDING DEVICE

### Abstract

An ink refill system with an ink storage bottle in which, when ink is supplied from an ink storage bottle **20** to a storage chamber **100** of a liquid tank **16,** the flow path resistance of a second flow path portion **192** through which the ink flows is less than the flow path resistance of a first flow path portion **191** through which air flows of the two flow path portions provided in a nozzle **110** of the ink storage bottle **20.**

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### Background/Summary

## BACKGROUND

### Field

[0001] The present disclosure relates to a recording device including an ink storage bottle that stores ink and a liquid tank to which the ink storage bottle is connected.

### Description of the Related Art

[0002] Some liquid tanks used in liquid discharge devices, such as ink jet recording devices, can be supplied with liquids from separately prepared ink storage bottles. In such a liquid discharge device, the ink storage bottle is attached to the liquid tank when ink is supplied to the liquid tank. When the ink storage bottle is attached to the liquid tank, the interior of the ink storage bottle communicates with the interior of the liquid tank through a flow path provided in the ink storage bottle. Printing liquids, such as ink, can be supplied from the ink storage bottle to the liquid tank by pressing the side wall of the ink storage unit of the ink storage bottle with the ink storage bottle attached to the liquid tank.

[0003] In Japanese Patent Laid-Open No. 2020-189454, ink is supplied from the ink storage bottle to the liquid tank by using a so-called chicken feed method. The ink storage bottle has two flow paths through which a liquid or air flows, and gas-liquid exchange can be performed by causing air to flow through one flow path and ink to flow through the other flow path. The ink level in the liquid tank rises when the ink flows into the liquid tank, and the air flow is blocked when the ink level reaches the opening of the flow path through which air flows in the ink storage bottle. As a result, the ink flow from the ink storage bottle to the liquid tank is stopped.

[0004] In the chicken feed method described above, after the ink storage bottle is attached to the liquid tank, the flow rate of supplying ink from the ink storage bottle to the liquid tank may be insufficient, and accordingly, the refill time becomes longer.

[0005] In Japanese Patent Laid-Open No. 2020-189454, the supply of ink from the ink storage bottle to the liquid tank is performed by gas-liquid exchange through the two flow paths provided in the nozzle of the ink storage bottle. The lengths of the two flow paths in the direction of a liquid flow are identical to each other, and the cross-sectional shapes and the cross-sectional areas are also identical to each other.

[0006] In addition, since the nozzle has a shape that projects from the storage unit of the ink storage bottle, the two flow paths are long tubes. As a result, there is a concern that the amount of ink supplied from the ink storage bottle to the liquid tank per unit time reduces, and accordingly, the refill time becomes longer.

## SUMMARY

[0007] The present disclosure addresses the problems described above and increases the refill speed of the ink storage bottle that supplies ink to the liquid tank and reduces the refill time.

[0008] Some exemplary embodiments, features, and aspects of the present disclosure have the following structure.

[0009] A liquid storage bottle that supplies a liquid to a recording device including a recording head that discharges the liquid, and a liquid tank that includes a storage unit that stores the liquid supplied to the recording head and an opening portion through which the liquid is supplied. The liquid storage bottle includes [0010] a bottle main body that stores the liquid; [0011] and a liquid supply unit including a second flow path through which the liquid flows from the bottle main body to the liquid tank, the second flow path being connected to the opening portion of the liquid tank, and a first flow path through which a gas flows from the liquid tank to the bottle main body, [0012] in which a flow path resistance of the first flow path is greater than a flow path resistance of the second flow path.

[0013] According to another exemplary embodiment, an ink refill system has an ink storage bottle in which, when ink is supplied from an ink storage bottle to a storage chamber of a liquid tank, the flow path resistance of a second flow path portion through which the ink flows is less than the flow

path resistance of a first flow path portion through which air flows of the two flow path portions provided in a nozzle of the ink storage bottle.
[0014] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0015] FIG. **1** is a perspective view of a recording device according to a first embodiment.
[0016] FIG. **2** is a side view illustrating a main portion of the recording device according to the first embodiment.
[0017] FIG. **3** is a perspective view illustrating a state in which a liquid is supplied to the recording device illustrated in FIG. **1**.
[0018] FIG. **4** is a diagram illustrating the component structure of an ink storage bottle according to the first embodiment.
[0019] FIG. **5** is a cross-sectional view of a nozzle main body illustrated in FIG. **4**.
[0020] FIG. **6** is a cross-sectional view illustrating a state in which the liquid is supplied in the first embodiment.
[0021] FIGS. **7**A to **7**D are cross-sectional views of the nozzle main body according to the first embodiment.
[0022] FIGS. **8**A to **8**L are plan views of a base end portion of a nozzle according to the first embodiment.
[0023] FIGS. **9**A to **9**D are plan views of a base end portion of a nozzle according to a second embodiment.
[0024] FIGS. **10**A to **10**H are cross-sectional views of a nozzle main body according to a third embodiment.
[0025] FIGS. **11**A to **11**E are cross-sectional views of a nozzle main body according to a fourth embodiment.
[0026] FIGS. **12**A to **12**D are cross-sectional views of the nozzle main body according to the fourth embodiment.
[0027] FIGS. **13**A to **13**H are front views of the nozzle main body according to the fourth embodiment.
[0028] FIGS. **14**A and **14**B are cross-sectional views of a nozzle main body according to a fifth embodiment.
[0029] FIGS. **15**A to **15**C are front views of the nozzle main body according to the fifth embodiment.
[0030] FIGS. **16**A to **16**D are front views of the leading end of a nozzle according to the fifth embodiment.
[0031] FIGS. **17**A to **17**G are cross-sectional views of a nozzle main body according to a sixth embodiment.
[0032] FIG. **18**A illustrates a plan view of the leading end of a nozzle according to the sixth embodiment, and FIGS. **18**B and **18**C illustrate a front view of a nozzle main body according to the sixth embodiment.
[0033] FIGS. **19**A, **19**C, **19**E, and **19**G are plan views of the leading end of a nozzle according to a seventh embodiment and FIGS. **19**B, **19**D, **19**F, and **19**H illustrate the shapes of openings of an injection portion according to the seventh embodiment.
[0034] FIGS. **20**A to **20**F are plan views of the leading end of a nozzle according to an eighth embodiment.
DESCRIPTION OF THE EMBODIMENTS

[0035] Various exemplary embodiments, features, and aspects of the present disclosure will be described in detail below with reference to the drawings. However, the dimensions, the materials, the shapes, the relative arrangements, and the like of components described in the present embodiment are not intended to limit the scope of the present disclosure to only those unless otherwise specified.
[0036] In addition, unless otherwise specified, the materials and the shapes of the components previously described in the text will remain the same in subsequent descriptions.
Structure of Device
[0037] FIG. **1** is a perspective view of an ink jet recording device according to an embodiment of the present disclosure. FIG. **2** is a side view schematically illustrating a main portion of the ink jet recording device according to the present embodiment.
[0038] An ink jet recording device **1000**, which is also referred to below as a recording device, includes a first feeding unit **1**, a second feeding unit **2**, a recording unit **3**, and a liquid supply unit **4**. The first feeding unit **1** includes a feeding roller **10** that separates a recording medium from a bundle of loaded recording media one by one and supplies the separated medium to the second feeding unit **2**. The second feeding unit **2** includes a conveying roller **11**, provided downstream of the first feeding unit **1** in the conveying direction of the recording media, that conveys the recording medium fed from the feeding roller **10**, and a paper discharge roller **12**. A platen **13** that supports, from below, the recording medium conveyed by the second feeding unit **2** is provided between the conveying roller **11** and the paper discharge roller **12**.
[0039] The recording unit **3** includes a carriage **14**, provided at a position facing the platen **13**, that moves reciprocally in a direction orthogonal to the conveying direction of the recording media and a recording head **15**, mounted on the carriage **14**, that has a plurality of rows of discharge ports each having a plurality of discharge ports. The recording head **15** discharges ink of different colors from the individual discharge port rows by energy generating elements corresponding to the discharge ports being driven in accordance with recording data and records a color image on the recording medium supported by the platen **13**.
[0040] The liquid supply unit **4** includes a liquid tank **16**, which is a semi-transparent or transparent container, and a flexible supply tube **107** that connects the liquid tank **16** and the recording head **15** to each other. In the present embodiment, four colors of ink (for example, cyan, magenta, yellow, and black) are used as the liquid, and four liquid tanks **16***a* to **16***d* that store the four colors of ink are provided as the liquid tank **16**. The liquid is not limited to ink and may also be a recording liquid, a fixing solution, a resist, or the like.
[0041] The liquid tank **16** includes a cap **40**, attachable to the tank body **160**, that hermetically seals the tank body **160** incorporating a storage chamber **100** that stores the liquid and the storage chamber **100**. A supply port **101** connected to the supply tube **107** is provided in a lower portion of the tank body **160**, and an atmospheric communication port **102** through which the storage chamber **100** communicates with the atmosphere is provided in the upper surface of the tank body **160**. When the liquid is discharged from the recording head **15**, the negative pressure in the recording head **15** increases, and accordingly, the liquid stored in the storage chamber **100** of the liquid tank **16** is supplied from the supply port **101** to the recording head **15** through the supply tube **107**. At this time, the same amount of gas as the liquid supplied to the recording head **15** flows into the storage chamber **100** in the liquid tank **16** through the atmospheric communication port **102**.
[0042] FIG. **3** is a perspective view illustrating a state in which the liquid is supplied to the recording device illustrated in FIG. **1**. When the remaining amount of the liquid in the storage chamber **100** is detected to be equal to or smaller than a certain amount by the remaining amount detection unit (not illustrated) provided in the liquid tank **16**, an indication prompting the user to supply the liquid to the liquid tank **16** is displayed on a display unit **1001** of the recording device **1000**. The user tilts and opens a tank cover **1002** provided on the front surface of the recording device **1000** forward, removes the cap **40** attached to the liquid tank **16** to which the liquid is

supplied, and exposes an opening portion **18** as a receiving portion to which the liquid is supplied. The opening portion **18** has an inclined surface that is inclined with respect to the horizontal direction and the vertical direction with the recording device **1000** installed in a normal use state, and the opening portion **18** is formed in the inclined surface. Then, the liquid is supplied to the liquid tank **16** through the exposed opening portion **18** by use of the ink storage bottle **20** that stores the liquid to be supplied. It should be noted that a plurality of types (four types in the present embodiment) of ink storage bottles **20** may be prepared in advance depending on the number of colors of the liquid (ink) used. Color information of the stored liquid (ink) is indicated on each of the ink storage bottles **20**. The user selects an ink storage bottle **20** that stores the liquid to be supplied from the plurality of prepared ink storage bottles **20** in accordance with the indication on the display unit **1001** and the color information displayed on the liquid tank **16**.

Structure of Bottle

[0043] FIG. **4** is a diagram illustrating the component structure of the ink storage bottle **20**, which is a liquid container for supplying a liquid to the liquid tank **16**. The ink storage bottle **20** includes a bottle main body **21** that stores the liquid, a nozzle main body **22** connected to the bottle main body **21**, and a bottle cap **23** attached to the nozzle main body **22**. The nozzle main body **22** has the function of serving as an outlet through which the liquid stored in the bottle main body **21** is discharged to the outside. The bottle cap **23** shields the inside of the ink storage bottle **20** from the outside by being attached to the nozzle main body **22**. The nozzle main body **22** is attachable to and detachable from the bottle main body **21** and is connected with a screwed manner. It should be noted that the nozzle main body **22** may be formed integrally with the bottle main body **21**. In the integral structure, for example, it is possible to use a method that seals them with a flexible component therebetween or a method that melts and connects the bottle main body **21** and the nozzle main body **22** made of a resin. The bottle cap **23** is a single component and is cylindrical in shape. The bottle cap **23** is attachable to and detachable from the nozzle main body **22**. The bottle cap **23** is attachable to and detachable from the bottle main body **21** via the nozzle main body **22**.

Structure of Nozzle

[0044] FIG. **5** is a cross-sectional view of the nozzle main body **22**. As illustrated in FIG. **5**, the nozzle **110** projects to a first orientation side **134** from the outer surface of a bottom wall **111** of a base end portion of the nozzle main body **22**. That is, the nozzle **110** projects to the first orientation side **134** from the bottle main body **21** through the nozzle main body **22** with the nozzle main body **22** attached to the bottle main body **21**. The nozzle **110** may project to a second orientation side **135** from the bottom wall **111** while projecting to the first orientation side **134** from the bottom wall **111**. In this case, the nozzle is provided so as to pass through the bottom wall **111** of the base end portion.

[0045] The nozzle **110** is substantially cylindrical in shape. The nozzle **110** has an outer circumferential surface having an annular cross section. A portion of the outer circumferential surface is tapered in shape and inclined in a direction in which the diameter of the outer circumferential circle decreases toward the leading end portion **115** of the nozzle **110** from the bottom wall **111**. Such a shape enables the nozzle **110** to be smoothly inserted into the tank sequentially from the distal end portion away from the bottle main body **21**. However, the outer circumferential surface of the nozzle **110** may extend vertically so as to have the same diameter of the outer circumferential surface from the bottom wall **111** to the leading end. Alternatively, the nozzle may also be, for example, square cylindrical in shape instead of cylindrical.

[0046] The nozzle **110** has a first flow path **191** and a second flow path **192** as flow paths **190** through which ink or air flows. The first flow path **191** and the second flow path **192** pass through the nozzle main body **22** along the first orientation side **134**. The first flow path **191** and the second flow path **192** extend to the first orientation side **134** but may also be curved. The lengths of the first flow path **191** and the second flow path **192** in the direction of an ink flow may be substantially the same or be different. In addition, the cross-sectional shapes and the cross-sectional areas of the first flow path **191** and the second flow path **192** may be substantially the same or be different. Furthermore, the number of the flow paths **190** may be more than two. The plurality of flow paths **190** may have different lengths or shapes. In a state in which the nozzle main body **22** is attached to the bottle main body **21**, one ends of the flow paths **190** are the base end portion of the nozzle **110** and communicate with the bottle main body **21** through a first opening **193** and a third opening **195**. The other ends of the flow paths **190** are the leading end portion of the nozzle **110** and communicate with the outside of the nozzle main body **22** through the second opening **194** and the fourth opening **196**. That is, the first flow path has a space volume with the first opening **193** and the second opening **194** at both ends, and the second flow path has a space volume with the third opening **195** and the fourth opening **196** at both ends. The first opening **193** and the third opening **195** are formed on the same plane at the base end portion. However, the first opening **193** and the third opening **195** may be formed on different surfaces. The second opening **194** and the fourth opening **196** are formed at the leading end portion **115** of the nozzle **110**.

[0047] The second opening **194** and the fourth opening **196** are circular in shape. It should be noted that the second opening **194** and the fourth opening **196** may have a shape other than a circle.

[0048] The leading end portion **115** of the nozzle **110** is a portion of the nozzle **110** formed of, for example, the leading end surface and the outer circumferential surface. The nozzle **110** has a recessed portion **116** in the outer circumferential surface thereof. The recessed portion **116** is defined by the leading end surface and an inner circumferential surface **118** (one of the side surfaces) of a circular rib **117** that projects to the first orientation side **134** from the outer edge of the leading end surface. That is, the leading end surface is recessed from the leading end (the leading end of the circular rib **117**) of the nozzle **110**. The inner circumferential surface **118** extends toward the outer edge of the leading end surface with distance from the leading end surface to the first orientation side **134**. That is, the inner circumferential surface **118** extends to the first orientation side **134** while being inclined in a direction that increases the diameter of the recessed portion **116**. It should be noted that the inner circumferential surface **118** may also extend to the first orientation side **134** without being inclined. In addition, the nozzle **110** does not need to have the recessed portion **116**. That is, the leading end portion of the nozzle **110** does not need to be recessed.

Supplying Ink

[0049] As illustrated in FIG. **6**, the nozzle **110** of the ink storage bottle **20** is inserted into the opening portion **18**, from which the liquid is injected, of the liquid tank **16**, and accordingly, the ink storage bottle **20** is connected to the liquid tank **16**. The posture of the ink storage bottle **20** with the ink storage bottle **20** connected to the liquid tank **16** is also referred to as the connection posture. The posture of the ink storage bottle **20** can be defined such that a mark **24** (see FIG. **4**) on the nozzle main body **22** faces vertically upward side when the nozzle **110** of the ink storage bottle **20** is inserted into the opening portion **18** of the liquid tank **16**.

[0050] That is, when the second opening **194** and the fourth opening **196** are located in the storage chamber **100**, one of the second opening **194** and the fourth opening **196** can be located vertically above the other. In the present embodiment, the second opening **194** is located above the fourth opening **196** in the connection posture. When the ink storage bottle **20** is connected to the liquid tank **16** and the second opening **194** and the fourth opening **196** are located in the storage chamber **100** of the liquid tank **16**, the bottle main body **21** communicates with the storage chamber **100** through the flow path **190**. In the connection posture in FIG. **6**, ink is supplied from the ink storage bottle **20** to the liquid tank **16** by a so-called chicken feed method.

[0051] That is, when the ink storage bottle **20** is connected to the liquid tank **16** and the second opening **194** and the fourth opening **196** are located in the storage chamber **100** of the liquid tank **16**, the bottle main body **21** communicates with the storage chamber **100** through the first flow path **191** and the second flow path **192**. As a result, the ink stored in the bottle main body **21** flows into the second flow path **192** through the third opening **195** and then flows into the storage chamber

**100** through the fourth opening **196**.

[0052] In addition, as the ink flows, air enters the storage chamber **100** through the atmospheric communication port **102** and flows into the bottle main body **21** through the first flow path **191**. Here, the volume of the ink flowing into the storage chamber **100** from the bottle main body **21** is substantially the same as the volume of air flowing into the bottle main body **21** from the storage chamber **100**. As described above, gas-liquid exchange is performed. When the ink flows into storage chamber **100** and the ink level in the storage chamber **100** rises to the height of the opening **196**, that is, the ink level reaches the height (dashed line position P) of a marker **103**, a flow of air between storage chamber **100** and bottle main body **21** through first flow path **191** is blocked. Accordingly, the flow of the ink from the bottle main body **21** to the storage chamber **100** is stopped.

[0053] The ink can also be supplied from the ink storage bottle **20** to the storage chamber **100** by using a method that deforms the side wall **121** of the bottle main body **21** of the ink storage bottle **20** in addition to the chicken feed method described above.

First Embodiment

[0054] As described earlier, the first flow path **191** and the second flow path **192** in the nozzle main body **22** of the ink storage bottle **20** are long tubes. When the cross-sectional areas of the first flow path **191** and the second flow path **192** are the same with each other, the ink refill speed may become slower when the ink is supplied from the ink storage bottle **20** to the liquid tank **16**. As a result, there is a concern that it takes time to supply the ink to the liquid tank **16**.

[0055] On the other hand, in the present embodiment, of the two flow paths provided in the nozzle **110** of the ink storage bottle **20**, the flow path resistance of one flow path portion is smaller than that of the other flow path portion, as illustrated in the cross-sectional views in FIGS. **7**A to **7**D. The flow path resistance refers to a frictional force caused in a direction opposite to the flow between the liquid and the inner wall of the flow path and is mainly determined by the shape. The flow path resistance is proportional to the length in the direction in which the liquid flows and inversely proportional to the cross-sectional area of the flow path. The space volume of one flow path is made wider than that of the other flow path to create a difference in the flow path resistance.

[0056] Specifically, the space volume of the second flow path **192** located on the downward side in the vertical direction is set to be larger than the space volume of the first flow path **191**. The space volumes of the first flow path **191** and the second flow path **192** can be set to desirable volumes by adjusting the position of the partition wall **119** that separates the first flow path **191** from the second flow path **192** and the thickness of the outer circumferential wall **113**. The third opening **195** is wider than the first opening **193** in the second flow path **192**, and the space volume communicating with the base end portion **114** of the flow path is larger. As a result, a larger amount of ink flows into the second flow path **192** through the third opening **195** when the ink storage bottle **20** is tilted to the ink refill posture, and accordingly, the ink refill speed is improved. In addition, the opening area of the third opening **195** that communicates with the ink storage bottle **20** is larger than the opening area of the first opening **193**, and the space volume that communicates with the base end portion **114** is larger. The flow path portion is a long tube that extends from the base end portion **114** to the leading end portion **115**. When the ink flows from the third opening **195** toward the fourth opening **196** to supply the ink, the momentum of ink flowing from the ink storage bottle **20** to the third opening **195** applies a force for pushing ink from the base end portion **114** toward the leading end portion **115**. As a result, the ink refill speed can be further improved.

[0057] As illustrated in FIGS. **7**A and **7**B, the partition wall **119** that separates the first flow path **191** from the second flow path **192** can be provided. That is, the third opening **195** of the second flow path **192** can be made wider than the first opening **193** by the partition wall **119** being shifted toward the first flow path, that is, in the first direction **137**, with respect to the center axis **136** of the nozzle **110**. The partition wall **119** may be provided vertically in parallel to the center axis **136** or may be provided so as to be inclined with respect to the center axis **136** to make the third

opening **195** as wide as possible.
[0058] In FIG. **7**C, the space volume of the second flow path **192** is made larger than the space volume of the first flow path **191** by adjusting the thickness of the outer circumferential wall **113** that constitutes the first flow path **191**. In FIG. **7**D, the thickness of partition wall **119** that separates the first flow path **191** from the second flow path **192** is changed to make the space volume of the second flow path **192** larger than the space volume of the first flow path **191**, thereby making a difference between the space volumes of the two flow paths.
[0059] The difference between the flow path resistances of the two flow paths, which are the first flow path **191** and the second flow path **192**, preferably significantly differs from each other for sufficient improvement of the ink refill speed, which is the effect of the present disclosure, when the ink is injected. However, the opening areas of the first opening **193** and the second opening **194** provided at both ends of the first flow path **191** are preferably large enough to cause air to flow into the bottle main body **21**. Accordingly, the difference between the flow path resistances of the two flow paths can be 1.2 and 10 times. The difference between the flow path resistances of the two flow paths can be 2 to 5 times to efficiently dispose the flow paths in the nozzle main body **22**.
[0060] Examples of the shapes of openings of the two flow paths disposed at the base end portion **114** are illustrated in FIGS. **8**A to **8**L. The first opening **193** and the third opening **195** may have the same shape and different sizes as illustrated in FIGS. **8**A to **8**E and FIG. **8**K or may have different shapes as illustrated in FIGS. **8**F to **8**J and FIG. **8**L. The cross section of the liquid supply unit, orthogonal to the center axis **136** along the direction in which the liquid in the liquid supply unit flows, that passes through the center axis **136** can be divided into two regions with respect to the center line **139**. The opening portions are provided such that the area of the third opening **195** closer to the side of the second direction **138** is larger than the opening area of the first opening **193** closer to the side of the first direction **137** with reference to the center line **139**. The openings with different shapes can make the opening area of the third opening **195** larger and increase the amount of the ink flowing into the second flow path **192**. In addition, the first opening **193** at the base end portion **114** and the second opening **194** at the leading end portion **115**, as well as the third opening **195** at the base end portion **114** and the fourth opening **196** at the leading end portion **115** may have the same shape or different shapes that vary from the base end portion **114** toward the leading end portion **115**.
[0061] The mark **24** may be provided on the nozzle main body **22** to define the connection posture of the ink storage bottle **20** such that the nozzle **110** is inserted into the opening portion **18** to position the second flow path **192** on the downward side in the vertical direction of the first flow path **191**. As long as the operator can be guided to define the positional relationship between the first flow path **191** and the second flow path **192** in the connection posture, a method other than this may be used.
Second Embodiment
[0062] FIGS. **9**A to **9**D illustrate examples of the shapes of openings disposed at the base end portion **114** when the plurality of first flow paths **191** and the plurality of second flow paths **192** are present. When a plurality of flow paths is present, these flow paths are disposed such that the total space volume of the plurality of second flow paths through which the liquid flows is larger than the total space volume of the plurality of first flow paths **191** through which the gas flows. The opening portions are provided such that the total opening area of the third opening **195** closer to the side of the second direction **138** is larger than the total area of the first openings **193** closer to the side of the first direction **137** with reference to the center line **139**. Since the plurality of first flow paths **191** through which the gas flows is present in FIG. **9**A, the plurality of first openings **193** is provided. Since the diameter of the cross-sectional area of the first flow path **191** through which the gas flows is smaller than that of the second flow path **192**, the ink cannot be supplied if the liquid enters and blocks the flow path. Since the plurality of first flow paths **191** is present, even if one of them is blocked, the ink can be supplied because the gas flows through another flow path. Since the

plurality of first flow paths **191** and the plurality of second flow paths **192** are present in FIGS. **9**B and **9**C, the plurality of first openings **193** and the plurality of third openings **195** are present. Even if a flow path through which the gas or the liquid flows is blocked at one position, the ink can be supplied because gas-liquid exchange can be performed in a remaining flow path. Since the plurality of second flow paths **192** through which the liquid flows is present in FIG. **9**D, the plurality of third openings **195** is provided. Even if one of the second flow paths **192** through which the liquid flows is blocked, the liquid can be supplied by using another flow path.

Third Embodiment

[0063] FIGS. **10**A to **10**H illustrate another embodiment in which the position of the partition wall **119** that separates the first flow path **191** from the second flow path **192** and the thickness of the outer circumferential wall **113** are adjusted to make the space volume of the second flow path **192** larger than the space volume of the first flow path **191**. Specifically, the cross-sectional area of the flow path portion can be partially reduced. A step is provided between the first opening **193** and the second opening **194** of the first flow path **191**. A slope **142** may be formed instead of the step. In FIGS. **10**A to **10**D, the opening area of the second opening **194** provided closer to the leading end portion **115** of the nozzle **110** is smaller than that of the fourth opening **196**. Since the amount of air taken in at the start of gas-liquid exchange is small to supply ink, the ink is supplied gently. Once gas-liquid exchange begins, since the flow speed of air is high, air is transported to the ink storage bottle **20** through the first flow path **191**. In the second flow path **192**, the third opening **195** at the base end portion **114** is wider than the fourth opening **196** at the leading end portion **115**. Accordingly, the momentum of the ink flowing from the ink storage bottle **20** into the second flow path **192** makes the ink refill speed faster. At the moment at which supply of ink from ink storage bottle **20** starts, the ink is discharged gently, the ink can be suppressed from adhering to the nozzle **110**. In FIGS. **10**E to **10**H, the opening area of the first opening **193** closer to the base end portion **114** of the nozzle **110** is smaller than that of the third opening **195**. Since the opening area of the first opening **193** located on the first orientation side **134** is small while the ink storage bottle **20** is tilted to the refill position, the ink is less likely to be injected through the first opening **193**. Even if the ink is injected, the amount is small. Even if the injected ink is formed as a liquid film near the first opening **193**, since air flows through the second opening **194**, which has a larger than the first opening **193**, after gas-liquid exchange starts, an ink film can be easily broken. As a result, gas-liquid exchange is smoothly started. When the opening area of the first opening **193** is adjusted by the thickness of the partition wall **119** as illustrated in FIGS. **10**B and **10**D, the first opening **193** can be disposed at a high position closer to the first orientation side **134** when the ink storage bottle **20** is tilted to a storage posture.

Fourth Embodiment

[0064] FIGS. **11**A to **11**E illustrate examples of an embodiment in which the space volume of the first flow path **191** is smaller than the space volume of the second flow path **192**. Specifically, as illustrated in FIG. **11**A, the partition wall that separates the first flow path **191** from the second flow path **192** is disposed so as to be aligned with the center axis **136** along a direction in which the liquid in the liquid supply unit flows, but the outer circumferential wall **113** that constitutes the first flow path **191** is brought closer to the center axis **136**. Alternatively, the flow path is narrowed by a portion of the outer circumferential wall **113** that constitutes the first flow path **191** being brought closer in the direction of the center axis **136**. In FIGS. **11**B and **11**C, the first flow path **191** is narrowed toward the leading end portion **115** of the nozzle **110** with a step or an inflection point set as a reduction starting point **141** of the flow path. In FIGS. **11**D and **11**E, the first flow path **191** is narrowed toward the base end portion **114** of the nozzle **110** with a step or an inflection point set as the reduction starting point **141** of the flow path. By denting at least a portion of the nozzle **110** to the inside, at least a portion of the inner wall of the first flow path can be expanded into the space of the first flow path. As a result, the space volume of the second flow path **192** becomes larger than the space volume of the first flow path **191**. As a result, it is possible to make a difference between the space volumes of the first flow path **191** and the second flow path **192** without significantly deforming the outer shape of nozzle **110**.

[0065] FIGS. **12**A to **12**D illustrate examples of cross-sectional views of the nozzle **110** exemplified in FIGS. **11**A to **11**E. A recessed portion **165** formed by denting at least a portion of the nozzle **110** to the inside may have a shape formed by partially cutting out a circle on the cross-section of the nozzle **110**, as illustrated in FIGS. **12**A and **12**B. As illustrated in FIGS. **12**C and **12**D, the recessed portion **165** may have a notched shape. The notched shape is defined by the width W and the distance L toward the center line **139** of nozzle **110**. The notched shape may be a rectangle, a triangle, a trapezoid, a semicircle, an ellipse, or the like but is not limited to these shapes as long as the notched shape is formed by at least a portion thereof is internally recessed. The recessed portion **165** can be disposed on the circumference of the outer circumferential surface within a range of ±90° with the second direction 138 set as the starting point. The recessed portion 165 can be disposed within a range of ±45° with the first direction **137** set as the starting point.

[0066] FIGS. **13**A to **13**H are examples of front views of the nozzle **110** having the recessed portion **165** when the first flow path **191** faces the front. The recessed portion **165** is defined by combination of the width W, the length L of the recess, and heights H**1** and H**2**. Specifically, the shape is obtained by removing a portion of a structure, such as a cuboid, a cylinder, a triangular pyramid, a quadrangular pyramid, a cone, or a quadrangular truncated pyramid, from the first flow path **191**. The recessed portion **165** is not limited to these as long as the space volume of the second flow path **192** is larger than the space volume of the first flow path **191**. The recessed portion **165** described above may also be a portion of a connection portion that engages the opening portion of the liquid tank when the nozzle is connected to the opening portion to supply ink.

Fifth Embodiment

[0067] FIGS. **14**A and **14**B illustrate examples in which an overhanging projecting portion **150** projecting in the second direction **138** is provided on a portion of the outer circumferential wall **113** that constitutes the second flow path **192** to make the space volume of the second flow path **192** larger than the space volume of the first flow path **191**. The overhanging projecting portion **150** is disposed to project to the outside with a step illustrated in FIG. **14**A or an inflection point illustrated in FIG. **14**B set as a starting point **151** for expanding the flow path. FIGS. **15**A to **15**D illustrate examples of front views of the nozzle **110** having the projecting portion **150** when the second flow path **192** faces the front. FIGS. **16**A to **16**D are plan views of the nozzle **110** as seen from the leading end portion **115**. The projecting portion **150** is defined by the heights H**1** and H**2**, the lengths L**1**, L**2**, and L**5** of overhangs in the second direction **138** from the outer circumferential surface, and widths W**1** and W**2** of projections. Specifically, the shape is formed by combining a portion of a structure, such as a cuboid, a cylinder, a triangular pyramid, a quadrangular pyramid, a cone, or a quadrangular truncated pyramid, with the second flow path **192**. The projecting portion **150** is not limited to these as long as the space volume of the second flow path **192** is larger than the space volume of the first flow path **191**. The overhanging projecting portion **150** may also be a portion of the connection portion that engages the opening portion of the liquid tank when the nozzle is connected to the opening portion to supply ink. The projecting portion **150** can be disposed on the circumference of the outer circumferential surface within a range of ±90° with the second direction **138** set as the starting point. The projecting portion **150** is disposed within a range of ±45° with the second direction **138** set as the starting point. The opening area of the third opening **195** can be made wider by the projecting portion **150** being disposed in the second flow path **192**. As a result, the amount of ink flowing from the third opening **195** to the second flow path **192** can be increased in the ink refill posture, and accordingly, the ink refill speed can be increased.

Sixth Embodiment

[0068] As illustrated in FIGS. **17**A to **17**G, an additional projecting portion **155** that expands the first flow path **191** may be provided in addition to the projecting portion **150** that expands the second flow path **192**. When the ink storage bottle **20** takes the connection posture, the projecting portions **150** and **155** are disposed such that the second flow path **192** is located on the downward side in the vertical direction and the space volume of the second flow path **192** is larger than the space volume of the first flow path **191**. Similar to the projecting portion **150**, the additional projecting portion **155** is disposed to overhang to the outside with steps illustrated in FIGS. **17**A to **17**C or inflection points illustrated in FIGS. **17**D to **17**F set as starting points. The lengths of overhangs in the first direction **137** from the outer circumferential surface are defined as L**3**, L**4**, and L**6**. As illustrated in FIG. **17**G, the combination of structures in which the projecting portions **150** and **155** overhang at a step and an inflection point is allowed and the combination is not limited to this. FIGS. **18**A to **18**C illustrate an example of a plan view of the nozzle **110** as viewed from the leading end portion **115** and examples of front views of the additional projecting portion **155**. The additional projecting portion **155** is defined by combination of the heights H**3** and H**4** and widths W**3** and W**4** of the overhanging additional projecting portion **155**. Specifically, the shape is formed by combining a portion of a structure, such as a cuboid, a cylinder, a triangular pyramid, a quadrangular pyramid, a cone, or a quadrangular truncated pyramid with the first flow path **191** but not limited to this. The additional projecting portion **155** can be disposed on the circumference of the outer circumferential surface within a range of ±90° with the first direction **137** set as the starting point. The additional projecting portion **155** can be disposed within a range of ±45° with the first direction **137** set as the starting point.

Seventh Embodiment

[0069] The projecting portion **150** and the additional projecting portion **155** provided on the outer circumferential surface of the nozzle **110** may also serve as a positioning mechanism for connecting the ink storage bottle **20** to the storage chamber **100** of the liquid tank **16**. The projecting portion **150** and the additional projecting portion **155** are cuboids here, but the shape is not limited to this. As illustrated in FIGS. **19**A to **19**H, the opening **19** of the opening portion **18** of the storage chamber **100** to which the ink storage bottle **20** is connected has a shape that fits to the nozzle **110**. As a result, the ink refill posture can be defined by using the projecting portion **150** or both the projecting portion **150** and the additional projecting portion **155**. In the keyhole shape illustrated in FIGS. **19**A and **19**B, a pull-out hole **161** into which the projecting portion **150** is inserted is provided in the opening **19**.

[0070] As illustrated in FIGS. **19**C and **19**D, a gap between a plurality of projecting portions **150***a* and **150***b* may be fixed by clamping a projection **162** provided in the opening **19**. As illustrated in FIGS. **19**E and **19**F, a receiving portion **163** and an additional receiving portion **164** are provided in the opening portion **18** such that the projecting portion **150** and the additional projecting portion **155** can be fitted into and abut against the receiving portion **163** and the additional receiving portion **164**. As illustrated in FIGS. **19**G and **19**H, a convex portion **166** and the pull-out hole **161** may be provided in the opening portion **18** so as to engage the recessed portion **165** and the additional projecting portion **155**. Only one example is described here, the present disclosure is not limited to this, and any combination is enabled. When the connection posture of the ink storage bottle **20** is defined by the mark **24** on the nozzle main body **22**, the user may manually adjust the posture after visual confirmation. Since the connection posture is adjusted manually, the first flow path **191** may deviate from its normal position located in an upper portion in the vertical direction of the second flow path **192** due to misrecognition of the mark **24** or error in manual adjustment of the posture position. Since the shape of the nozzle **110** physically defines the ink refill posture of the ink storage bottle **20** in the present embodiment as illustrated in FIGS. **19**A to **19**H, the user can make connection to the storage chamber **100** at a correct position even when misrecognizing the mark **24**. In addition, since the second flow path **192** is always located on the downward side in the vertical direction by providing the projecting portion **150** to make the space volume of the second flow path **192** through which ink flows larger than the space volume of the first flow path **191**, a larger amount of ink flows from the ink storage bottle **20** to the second flow path **192**.

[0071] As a result, both definition of the ink refill posture and an increase in the ink refill speed can be achieved.

Eighth Embodiment

[0072] When there is a plurality of liquid tanks **16**, different combinations may be set in the storage chambers **100** such that the shapes of the projecting portions **150** and **155** and the recessed portions **165** provided in the ink storage bottle **20** fit to the shape of the opening **19** of the opening portion **18** of the storage chamber **100**. An example is illustrated in FIGS. **20**A to **20**E. When a plurality of different types of ink is used, combination in which the shape of the projecting portion **150** fits to the shape of the opening **19** of the opening portion **18** of the storage chamber **100** can be changed for each type of ink. For example, it is possible to adopt a combination in which the position of the projecting portion **150** provided in the second flow path **192** is the same, but the position of the additional projecting portion **155** provided in the first flow path **191** is different, as illustrated in FIG. **20**A and FIG. **20**B. As illustrated in FIGS. **20**C and **20**D, the position of the additional projecting portion **155** provided in the first flow path **191** may be the same, and the combination of the shapes and arrangement of the projecting portions **150***a* and **150***b* provided in the second flow path **192** may be changed. As illustrated in FIGS. **20**E and **20**F, the recessed portion **165**, the projecting portions **150** (**150***a* and **150***b*), and the additional projecting portions **155** (**155***a* and **155***b*) may be freely combined with each other. As described above, by changing the combination of the projecting portions **150** and **155**, the recessed portion **165**, and the opening **19** for each liquid when the user supplies the ink from the ink storage bottle **20** to the storage chamber **100**, the user can avoid wrong insertion of the liquid storage bottle **200**, thereby improving operability.

[0073] The embodiments described above may be combined as appropriate.

[0074] According to the present disclosure, since the amount of liquid supplied from the ink storage bottle to the liquid tank per unit time increases, the refill speed is improved, and the refill time can be reduced.

[0075] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0076] This application claims the benefit of priority from Japanese Patent Application No. 2024-032737, filed Mar. 5, 2024, which is hereby incorporated by reference herein in its entirety.

## Claims

**1**. A liquid storage bottle that supplies a liquid to a recording device including a recording head that discharges the liquid, and a liquid tank that includes a storage unit that stores the liquid supplied to the recording head and an opening portion through which the liquid is suppled, the liquid storage bottle comprising: a bottle main body that stores the liquid; and a liquid supply unit including a second flow path through which the liquid flows from the bottle main body to the liquid tank, the second flow path being connected to the opening portion of the liquid tank, and a first flow path through which a gas flows from the liquid tank to the bottle main body, wherein a flow path resistance of the first flow path is greater than a flow path resistance of the second flow path.

**2**. The liquid storage bottle according to claim 1, wherein a flow path resistance of the first flow path is 1.2 to 10 times a flow path resistance of the second flow path.

**3**. The liquid storage bottle according to claim 2, wherein the flow path resistance of the first flow path is 2 to 5 times a flow path resistance of the second flow path.

**4**. The liquid storage bottle according to claim 1, wherein a length of the first flow path is substantially identical to a length of the second flow path, and a cross-sectional area of the second flow path is larger than a cross-sectional area of the first flow path.

**5**. The liquid storage bottle according to claim 1, wherein a connection posture is defined such that a projecting portion or a recessed portion of the liquid supply unit engages the opening portion and the first flow path is located vertically above the second flow path.

**6**. The liquid storage bottle according to claim 5, wherein the projecting portion or the recessed portion of the liquid supply unit varies depending on a type of the liquid.

**7**. The liquid storage bottle according to claim 5, wherein the liquid supply unit has a positioning indication that defines the connection posture.

**8**. The liquid storage bottle according to claim 1, wherein at least one of the first flow path and the second flow path includes a plurality of flow paths.

**9**. A recording device comprising: the liquid storage bottle according to claim 1; a recording head that performs recording by discharging a liquid; and a liquid tank, connected through the liquid supply unit, that stores the liquid injected from the liquid storage bottle.

**10**. The recording device according to claim 9, wherein the liquid tank has an inclined surface inclined in both a horizontal direction and a vertical direction, and an opening portion into which the liquid supply unit of the liquid storage bottle is inserted is formed in the inclined surface.

**11**. The recording device according to claim 9, wherein the liquid tank has an atmospheric communication port through which an inside of the liquid tank communicates with an atmosphere.

**12**. An ink refill system with an ink storage bottle in which, when ink is supplied from an ink storage bottle to a storage chamber of a liquid tank, the flow path resistance of a second flow path portion through which the ink flows is less than the flow path resistance of a first flow path portion through which air flows of the two flow path portions provided in a nozzle of the ink storage bottle.

**13**. The ink refill system according to claim 12, wherein the ink storage bottle comprises a liquid storage bottle that supplies a liquid to a recording device including a recording head that discharges the liquid, and a liquid tank that includes a storage unit that stores the liquid supplied to the recording head and an opening portion through which the liquid is suppled, the liquid storage bottle comprising: a bottle main body that stores the liquid; and a liquid supply unit including a second flow path through which the liquid flows from the bottle main body to the liquid tank, the second flow path being connected to the opening portion of the liquid tank, and a first flow path through which a gas flows from the liquid tank to the bottle main body, wherein a flow path resistance of the first flow path is greater than a flow path resistance of the second flow path.

**14**. The ink refill system according to claim 13, wherein a flow path resistance of the first flow path is 1.2 to 10 times a flow path resistance of the second flow path.

**15**. The ink refill system according to claim 14, wherein the flow path resistance of the first flow path is 2 to 5 times a flow path resistance of the second flow path.

**16**. The ink refill system according to claim 13, wherein a length of the first flow path is substantially identical to a length of the second flow path, and a cross-sectional area of the second flow path is larger than a cross-sectional area of the first flow path.

**17**. The ink refill system according to claim 13, wherein a connection posture is defined such that a projecting portion or a recessed portion of the liquid supply unit engages the opening portion and the first flow path is located vertically above the second flow path.

**18**. The ink refill system according to claim 17, wherein the projecting portion or the recessed portion of the liquid supply unit varies depending on a type of the liquid.

**19**. The ink refill system according to claim 17, wherein the liquid supply unit has a positioning indication that defines the connection posture.

**20**. The ink refill system according to claim 13, wherein at least one of the first flow path and the second flow path includes a plurality of flow paths.